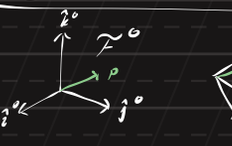
Chapter 2



$\hat{i}^i, \hat{j}^i, \hat{k}^i$ ($\cdot^i$ denotes frame)
Mutually orthogonal bases

There are different frames and we want to express a vector in one frame using the other

In $\mathcal{F}^0$ we have $\vec{p} = p_x^0 \hat{i}^0 + p_y^0 \hat{j}^0 + p_z^0 \hat{k}^0$
In $\mathcal{F}^1$ we have $\vec{p} = p_x^1 \hat{i}^1 + p_y^1 \hat{j}^1 + p_z^1 \hat{k}^1$

$$\Rightarrow p_x^0 \hat{i}^0 + p_y^0 \hat{j}^0 + p_z^0 \hat{k}^0 = p_x^1 \hat{i}^1 + p_y^1 \hat{j}^1 + p_z^1 \hat{k}^1$$

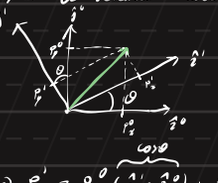
Doing a dot prod on each side with the bases of $\mathcal{F}^1$:

$$p_x^1 \underbrace{\hat{i}^1 \cdot \hat{i}^1}_{1} + p_y^1 \underbrace{\hat{j}^1 \cdot \hat{i}^1}_{0} + \dots = (p_x^0 \hat{i}^0 + \dots)\cdot \hat{i}^1$$

going from frame 0 to 1

$$\Rightarrow \vec{p}^1 = \begin{bmatrix} p_x^1 \\ p_y^1 \\ p_z^1 \end{bmatrix} = \begin{bmatrix} \hat{i}^1\cdot\hat{i}^0 & \hat{i}^1\cdot\hat{j}^0 & \hat{i}^1\cdot\hat{k}^0 \\ \hat{j}^1\cdot\hat{i}^0 & \hat{j}^1\cdot\hat{j}^0 & \hat{j}^1\cdot\hat{k}^0 \\ \hat{k}^1\cdot\hat{i}^0 & \hat{k}^1\cdot\hat{j}^0 & \hat{k}^1\cdot\hat{k}^0 \end{bmatrix} \begin{bmatrix} p_x^0 \\ p_y^0 \\ p_z^0 \end{bmatrix} \Rightarrow \vec{p}^1 = R_0^1 \vec{p}^0$$

For a rotation about $\hat{k}^0 = \hat{k}^1$:



$$\Rightarrow p_x^1 = p_x^0 \underbrace{(\hat{i}^1\cdot\hat{i}^0)}_{\cos\theta} + p_y^0 \underbrace{(\hat{i}^1\cdot\hat{j}^0)}_{\sin\theta}$$

$$p_y^1 = p_x^0 \underbrace{(\hat{j}^1\cdot\hat{i}^0)}_{-\sin\theta} + p_y^0 \underbrace{(\hat{j}^1\cdot\hat{j}^0)}_{\cos\theta}$$

Basically projecting the original basis onto the new basis gives the factors

Frames:

$\mathcal{F}^i$: inertial, earth axes — $\hat{j}$: East, $\hat{i}$: North, $\hat{k}$: centre of earth | NED

$\mathcal{F}^v$: Vehicle, coincides for pos. — $\hat{i}, \hat{j}, \hat{k}$ | aligned w NED

$\mathcal{F}^{v1}$: Vehicle frame 1, coincides for heading — rotation about $\hat{k}^v$ by heading $\Rightarrow \vec{p}^{v1} = R_v^{v1}(\psi)\vec{p}^v = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}\vec{p}^v$

$\mathcal{F}^{v2}$: Vehicle frame 2, coincides for pitch ... — rotation about $\hat{j}^{v2} \Rightarrow p^{v2} = R_{v1}^{v2}(\theta)\vec{p}^{v1} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}\vec{p}^{v1}$

$\mathcal{F}^b$: body, coincides for roll ... — rotation about $\hat{i}^{v2} \Rightarrow p^b = R_{v2}^b(\phi)\vec{p}^{v2} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix}\vec{p}^{v2}$

$$\Rightarrow \vec{p}^b = R_{v2}^b(\phi) R_{v1}^{v2}(\theta) R_v^{v1}(\psi)\vec{p}^v$$

$\mathcal{F}^s$: Stability, $\hat{i}$ is "aligned" with flow, for angle of attack

$\mathcal{F}^w$: wind, for sideslip

Velocity:

V_g – inertial velocity (ground vel)
V_a – Vel relative to air
V_w – wind velocity

$$\vec{V}_a = \vec{V}_g - \vec{V}_w$$

$$\Rightarrow V_g^b = \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$ inertial velocity in body frame

$$V_w^b = \begin{bmatrix} u_w \\ v_w \\ w_w \end{bmatrix} = R_v^b(\phi,\theta,\psi)\begin{pmatrix} w_n \\ w_e \\ w_d \end{pmatrix}$$ wind velocity in body frame

$$V_a^b = \begin{bmatrix} u_r \\ v_r \\ w_r \end{bmatrix} = \begin{bmatrix} u - u_w \\ v - v_w \\ w - w_w \end{bmatrix}$$ relative airspeed in body frame

From a similar analysis to above, using the definition of V_a and rotating to the body frame:

$$\begin{bmatrix} u_r \\ v_r \\ w_r \end{bmatrix} = V_a \begin{bmatrix} \cos\alpha\cos\beta \\ \sin\beta \\ \sin\alpha\cos\beta \end{bmatrix}$$ Very important for Physics

$$\Rightarrow V_A = \sqrt{u_r^2 + v_r^2 + w_r^2}$$

$$\alpha = \tan^{-1}\left(\frac{w_r}{u_r}\right)$$

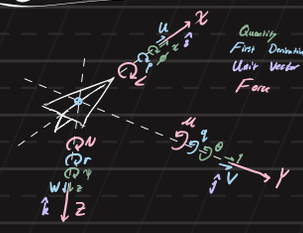
$$\beta = \sin^{-1}\left(\frac{v_r}{\sqrt{u_r^2+v_r^2+w_r^2}}\right)$$

Derivatives

$$\frac{d}{dt_i}\vec{p} = \frac{d}{dt_b}\vec{p} + \vec{\omega}_{b/i} \times \vec{p}$$ $\omega_{b/i}$: angular vel of body frame relative to inertial

1.) Set up simple rendering
2.) Implement rotation functions $\Rightarrow$ one function taking angles and outputting rotation matrix
3.) Test translation and rotation

Chapter 3



Kinematic relations

$$\frac{d}{dt}\begin{bmatrix} p_n \\ p_e \\ p_d \end{bmatrix} = R_b^v \begin{bmatrix} u \\ v \\ w \end{bmatrix} = (R_v^b)^T \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

each angle is in addition FOR so we diff in that FOR and then rotate

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} \dot\phi \\ 0 \\ 0 \end{bmatrix} + R_{v_1}^b(\phi)\begin{bmatrix} 0 \\ \dot\theta \\ 0 \end{bmatrix} + R_{v_2}^b(\phi) R_{v_1}^{v_2}(\theta)\begin{bmatrix} 0 \\ 0 \\ \dot\psi \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & -\sin\theta \\ 0 & \cos\phi & \sin\phi\cos\theta \\ 0 & -\sin\phi & \cos\phi\cos\theta \end{bmatrix}\begin{bmatrix} \dot\phi \\ \dot\theta \\ \dot\psi \end{bmatrix}$$

$$\begin{bmatrix} \dot p_n \\ \dot p_e \\ \dot p_d \end{bmatrix} = \begin{bmatrix} \cos\theta\cos\psi & \sin\phi\sin\theta\cos\psi - \cos\phi\sin\psi & \cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi \\ \cos\theta\sin\psi & \sin\phi\sin\theta\sin\psi + \cos\phi\cos\psi & \cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi \\ -\sin\theta & \sin\phi\cos\theta & \cos\phi\cos\theta \end{bmatrix}\begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

$$\begin{bmatrix} \dot\phi \\ \dot\theta \\ \dot\psi \end{bmatrix} = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi\sec\theta & \cos\phi\sec\theta \end{bmatrix}\begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

Dynamics

Translation

Using $V_g^b = [u\ v\ w]^T$, the velocity with respect to ground in the body frame

$$m\frac{d\vec{V}_g}{dt_i} = \underline{f} \quad \Big| \text{ we want in body frame} \Rightarrow = \frac{d\vec{V}_g}{dt_b} + \vec{\omega}_{b/i} \times \vec{V}_g$$

$\vec{\omega}_{b/i}$ from earlier is $[\dot\phi, \dot\theta, \dot\psi]^T$

Alternatively:

$$m\left(\frac{d\vec{V}_g^b}{dt_b} + \vec{\omega}_{b/i}^b \times \vec{V}_g^b\right) = \vec{f}^b \quad \text{which is in the body frame, } V_g^b = [u,v,w]^T,\ \omega_{b/i}^b = [p,q,r]$$

$$\frac{d\vec{V}_g}{dt_b} = \begin{bmatrix} \dot u \\ \dot v \\ \dot w \end{bmatrix} \Rightarrow \begin{bmatrix} \dot u \\ \dot v \\ \dot w \end{bmatrix} = \begin{bmatrix} rv - qw \\ pw - ru \\ qu - pv \end{bmatrix} + \frac{1}{m}\begin{bmatrix} f_x^b \\ f_y^b \\ f_z^b \end{bmatrix}$$

Rotational Motion

$\vec{L}$ ← angular momentum

$$\frac{d\vec{L}}{dt_i} = \vec{\tau} = \frac{d\vec{L}}{dt_b} + \vec{\omega}_{b/i} \times \vec{L} \quad \Big| \text{ Assumes moments taken about COM}$$

Expressing in body frame:

$$\frac{d\vec{L}^b}{dt_b} + \vec{\omega}_{b/i}^b \times \vec{L}^b = \vec{\tau}^b \quad \Big| \quad \underline{I} = \begin{bmatrix} I_x & I_{xy} & I_{xz} \\ & I_y & I_{yz} \\ & & I_z \end{bmatrix},\ \frac{dI}{dt} = 0$$

$$\vec{L}^b = \underline{I}^b \vec{\omega}_{b/i} \Rightarrow \frac{d\vec{L}^b}{dt_b} = \underline{I}\frac{d\omega_{b/i}^b}{dt_b}$$

$$\Rightarrow \underline{I}\frac{d\vec{\omega}_{b/i}^b}{dt_b} + \vec{\omega}_{b/i}^b \times (\underline{I}\,\vec{\omega}_{b/i}^b) = \vec{\tau}^b \quad \Big| \quad \dot{\vec{\omega}}_{b/i}^b = \begin{bmatrix} \dot p \\ \dot q \\ \dot r \end{bmatrix} \text{ and } \dot\omega_{b/i}^b = \underline{I}^{-1}\left[-\omega_{b/i}^b \times (\underline{I}\,\vec{\omega}_{b/i}^b) + \vec{\tau}^b\right]$$

Using Symmetry:

$$\underline{I} = \begin{bmatrix} I_x & 0 & -I_{xz} \\ 0 & I_y & 0 \\ -I_{xz} & 0 & I_z \end{bmatrix} \Rightarrow \underline{I}^{-1} = \begin{bmatrix} \frac{I_z}{\Gamma} & 0 & \frac{I_{xz}}{\Gamma} \\ 0 & \frac{1}{I_y} & 0 \\ \frac{I_{xz}}{\Gamma} & 0 & \frac{I_x}{\Gamma} \end{bmatrix} \quad \text{with } \Gamma = I_x I_z - I_{xz}^2$$

$$\Rightarrow \begin{bmatrix} \dot p \\ \dot q \\ \dot r \end{bmatrix} = \begin{bmatrix} \frac{I_z}{\Gamma} & 0 & \frac{I_{xz}}{\Gamma} \\ 0 & \frac{1}{I_y} & 0 \\ \frac{I_{xz}}{\Gamma} & 0 & \frac{I_x}{\Gamma} \end{bmatrix}\left(\begin{bmatrix} 0 & r & -q \\ -r & 0 & p \\ q & -p & 0 \end{bmatrix}\begin{bmatrix} I_x & 0 & -I_{xz} \\ 0 & I_y & 0 \\ -I_{xz} & 0 & I_z \end{bmatrix}\begin{bmatrix} p \\ q \\ r \end{bmatrix} + \begin{bmatrix} L \\ M \\ N \end{bmatrix}\right)$$

$$= \begin{bmatrix} \frac{I_z}{\Gamma} & 0 & \frac{I_{xz}}{\Gamma} \\ 0 & \frac{1}{I_y} & 0 \\ \frac{I_{xz}}{\Gamma} & 0 & \frac{I_x}{\Gamma} \end{bmatrix}\left(\begin{bmatrix} I_{xz}pq + (I_y - I_z)qr \\ I_{xz}(r^2 - p^2) + (I_z - I_x)pr \\ (I_x - I_y)pq - I_{xz}qr \end{bmatrix} + \begin{bmatrix} L \\ M \\ N \end{bmatrix}\right)$$

$$= \begin{bmatrix} \frac{I_{xz}(I_x - I_y + I_z)}{I_x I_z - I_{xz}^2}pq - \frac{I_z(I_z - I_y) + I_{xz}^2}{I_x I_z - I_{xz}^2}qr + \frac{I_z}{I_x I_z - I_{xz}^2}L + \frac{I_{xz}}{I_x I_z - I_{xz}^2}N \\ \frac{I_z - I_x}{I_y}pr - \frac{I_{xz}}{I_y}(p^2 - r^2) + \frac{M}{I_y} \\ \frac{(I_x - I_y)I_x + I_{xz}^2}{I_x I_z - I_{xz}^2}pq - \frac{I_{xz}(I_x - I_y + I_z)}{I_x I_z - I_{xz}^2}qr + \frac{I_{xz}}{I_x I_z - I_{xz}^2}L + \frac{I_x}{I_x I_z - I_{xz}^2}N \end{bmatrix}$$

$$= \begin{bmatrix} \frac{I_{xz}(I_x - I_y + I_z)}{I_x I_z - I_{xz}^2}pq - \frac{I_z(I_z - I_y) + I_{xz}^2}{I_x I_z - I_{xz}^2}qr \\ \frac{I_z - I_x}{I_y}pr - \frac{I_{xz}}{I_y}(p^2 - r^2) \\ \frac{(I_x - I_y)I_x + I_{xz}^2}{I_x I_z - I_{xz}^2}pq - \frac{I_{xz}(I_x - I_y + I_z)}{I_x I_z - I_{xz}^2}qr \end{bmatrix} + \begin{bmatrix} \frac{I_z}{I_x I_z - I_{xz}^2}L + \frac{I_{xz}}{I_x I_z - I_{xz}^2}N \\ \frac{M}{I_y} \\ \frac{I_{xz}}{I_x I_z - I_{xz}^2}L + \frac{I_x}{I_x I_z - I_{xz}^2}N \end{bmatrix}$$

1.) We can use $\begin{bmatrix} \dot p_n \\ \dot p_e \\ \dot p_d \end{bmatrix} = \dots$, $\begin{bmatrix} \dot u \\ \dot v \\ \dot w \end{bmatrix} = \dots$, $\begin{bmatrix} \dot\phi \\ \dot\theta \\ \dot\psi \end{bmatrix} = \dots$ and $\begin{bmatrix} \dot p \\ \dot q \\ \dot r \end{bmatrix} = \dots$ to represent our aircraft

2.) Implement the different formulations using both hand calcs and the full vector equations

Re-expressing with Quaternions

Quaternion $\in \mathbb{R}^4$, $\quad \underline{e} = e_0 + e_1\hat{i} + e_2\hat{j} + e_3\hat{k}$, $\quad \hat{i}^2 = \hat{j}^2 = \hat{k}^2 = \hat{i}\hat{j}\hat{k} = -1$ } from this $\hat{i}\hat{j} = \hat{k} = -\hat{j}\hat{i}$

$\Rightarrow$ equivalently $\quad \underline{e} = [e_0, \vec{e}]^T$, $\quad \vec{e} = [e_1, e_2, e_3]^T$

If you rotate by θ about an axis $\vec{v}$, the quaternion is related via:

$$e_0 = \cos\left(\frac{\theta}{2}\right), \quad \vec{e} = \vec{v}\sin\left(\frac{\theta}{2}\right) = \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}$$

A quaternion product is defined as:

$$\underline{a} \otimes \underline{b} = \begin{bmatrix} a_0 b_0 - \vec{a}\cdot\vec{b} \\ a_0\vec{b} + b_0\vec{a} + \vec{a}\times\vec{b} \end{bmatrix}, \text{ with } \underline{a} = [a_0, \vec{a}]^T, \ \underline{b} = [b_0, \vec{b}]^T$$

↳ makes it non commutative

To actually rotate a vector $\vec{r}$, it's expressed as $(0, \vec{r})$, and we translate between $\underline{e}$ and its conjugate $\underline{e}^* (= [e_0, -\vec{e}]^T)$ (because $\frac{\theta}{2}$ is its own inverse)

$$(0, \vec{r}^V) = \underline{e} \otimes (0, \vec{r}^B) \otimes \underline{e}^*, \quad (0, \vec{r}^B) = \underline{e}^* \otimes (0, \vec{r}^V) \otimes \underline{e}$$

$\Rightarrow \underline{e} \otimes (\cdot) \otimes \underline{e}^*$ links body components to vehicle components

$\underline{e}$ is the attitude of the body frame relative to vehicle (i.e. started without translation)

The rotation Matrix

Expanding:

$$\underline{e} \otimes (0, \vec{r}) \otimes \underline{e}^* = \left[-\vec{e}\cdot\vec{r}^B, \ e_0\vec{r}^B + \vec{e}\times\vec{r}^B\right]^T \otimes \left[e_0, -\vec{e}\right]^T = \vec{r}^V$$

$$= \left[-e_0(\vec{e}\cdot\vec{r}^B) + (e_0\vec{r} + \vec{e}\times\vec{r}^B)\cdot\vec{e}, \ (\vec{e}\cdot\vec{r})\vec{e} + e_0(e_0\vec{r} + \vec{e}\times\vec{r}^B) + (e_0\vec{r} + \vec{e}\times\vec{r}^B)\times(-\vec{e})\right]^T$$

and with $\vec{e}\times(\vec{e}\times\vec{r}^B) = \vec{e}(\vec{e}\cdot\vec{r}) - \vec{r}(\vec{e}\cdot\vec{e})$ (vector identity, very common when manipulating Navier-Stokes)

$$\Rightarrow \Big[\underbrace{-e_0(\vec{e}\cdot\vec{r})}_{\text{equal but opposite sign}} + \underbrace{e_0(\vec{r}\cdot\vec{e})}_{\text{sign}} + \underbrace{(\vec{e}\times\vec{r}^B)\cdot\vec{e}}_{\vec{e}\perp\vec{r}^B \text{ so dot} = 0}, \ (\vec{e}\cdot\vec{r})\vec{e} + e_0^2\vec{r} + \underbrace{e_0(\vec{e}\times\vec{r}^B)}_{\vec{r}\times\vec{e} = -\vec{e}\times\vec{r}} - \underbrace{e_0(\vec{r}^B\times\vec{e})}_{= 2e_0(\vec{e}\times\vec{r}^B)} - \underbrace{(\vec{e}\times\vec{r}^B)\times\vec{e}}_{= +\vec{e}\times(\vec{e}\times\vec{r}^B) = (\vec{e}\cdot\vec{r})\vec{e} - (\underbrace{\vec{e}\cdot\vec{e}}_{=1})\vec{r}}\Big]^T$$

$$\Rightarrow \vec{r}^V = \underbrace{(e_0^2 - \vec{e}^T\vec{e})}_{(e_0^2 - \vec{e}^2)I}\vec{r}^B + 2(\vec{e}^T\vec{r}^B)\vec{e} + 2e_0(\vec{e}\times\vec{r}^B)$$

repeated + commutativity $\Rightarrow 2(\vec{e}\vec{e}^T)\vec{r}^B$ $\quad \vec{e}\times\vec{r}^B = \begin{bmatrix} e_2 r_3 - e_3 r_2 \\ e_3 r_1 - e_1 r_3 \\ e_1 r_2 - e_2 r_1 \end{bmatrix} = \begin{bmatrix} 0 & -e_3 & e_2 \\ e_3 & 0 & -e_1 \\ -e_2 & e_1 & 0 \end{bmatrix}\vec{r}$

$$\Rightarrow R_B^V(\underline{e}) = \begin{bmatrix} e_0^2 + e_1^2 - e_2^2 - e_3^2 & 2(e_1e_2 - e_0e_3) & 2(e_1e_3 + e_0e_2) \\ 2(e_1e_2 + e_0e_3) & e_0^2 - e_1^2 + e_2^2 - e_3^2 & 2(e_2e_3 - e_0e_1) \\ 2(e_1e_3 - e_0e_2) & 2(e_2e_3 + e_0e_1) & e_0^2 - e_1^2 - e_2^2 + e_3^2 \end{bmatrix} \Rightarrow R_V^B(\underline{e})^T$$

Quaternion Kinematics

over interval dt, body will turn by angle $\|\vec{\omega}\|dt$ about axis $\frac{\vec{\omega}}{\|\vec{\omega}\|}$ (in body axes as $\vec{\omega} = [p, q, r]^T$)

With $\underline{e} = [\cos\frac{\theta}{2}, \ \vec{v}\sin(\frac{\theta}{2})]$

$\Rightarrow \delta\underline{e} \approx [1, \ \frac{1}{2}\vec{\omega}\,dt]$

Rotating about current body axes will compose on the right

$\Rightarrow \underline{e}(t + dt) = \underline{e}(t) \otimes \delta\underline{e} = e(t) + e(t) \otimes (0, \frac{1}{2}\vec{\omega})dt$

$\Rightarrow \dot{\underline{e}} = \frac{1}{2}\underline{e} \otimes (0, \vec{\omega}) = \frac{1}{2}[0 - \vec{e}\cdot\vec{\omega}, \ e_0\vec{\omega} + \vec{e}\times\vec{\omega}]$

$$\Rightarrow \begin{bmatrix} \dot{e}_0 \\ \dot{e}_1 \\ \dot{e}_2 \\ \dot{e}_3 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 0 & -p & -q & -r \\ p & 0 & r & -q \\ q & -r & 0 & p \\ r & q & -p & 0 \end{bmatrix}\begin{bmatrix} e_0 \\ e_1 \\ e_2 \\ e_3 \end{bmatrix}$$

↑ the magnitude will change, so we need to renormalize (or do some quaternion discount Lorentz)

To sum up

The state is composed of:

- position (inertial frame) $= [P_N, P_E, P_d]^T = \vec{P}$
- ground velocity (body frame) $= [u, v, w]^T = \vec{V}_b$
- attitude (from body to vehicle) $= [e_0, e_1, e_2, e_3]^T = \underline{e} = [e_0, \vec{e}]$, $\vec{e} = [e_1\ e_2\ e_3]^T$
- rates (body frame) $= [p, q, r]^T = \vec{\omega}_b$

For forces and moments on the coast, we use body frame (usually)

1.) Calculate the rate of change of velocity and rates in the body frame
2.) Use the kinematic relations to express the position in the vehicle frame using integrated body frame velocity, and to update the attitude based on the body-frame rates.

Dynamics (body-frame)

$$\dot{\vec{V}}_b = -\vec{\omega}_b \times \vec{V}_b + \frac{1}{m}\vec{F}_b$$

$$\dot{\vec{\omega}}_b = \underline{I}^{-1}(\vec{\tau}_b - \vec{\omega}_b \times \underline{I}\vec{\omega}_b)$$

Kinematic (body to vehicle frame)

$$\dot{\vec{P}} = \underline{R}_b^V(\underline{e})\,\vec{V}_b$$

$$\dot{\underline{e}} = \frac{1}{2}\left(\underline{e} \otimes [0, \vec{\omega}_b]\right)$$

Forces and Moments

Forces and moments can be split (aircraft also sometimes include an augmental mass matrix)

$$\vec{f}_b = \underbrace{\vec{f}_g}_{\text{gravity}} + \underbrace{\vec{f}_a}_{\text{aero}} + \underbrace{\vec{f}_p}_{\text{propulsive}}$$

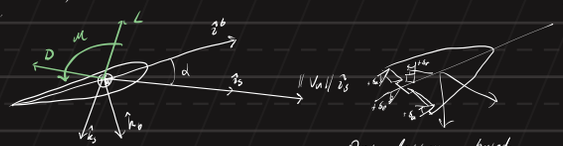
$$\vec{\tau}_b = \underbrace{\vec{\tau}_a}_{\text{aero}} + \underbrace{\vec{\tau}_p}_{\text{propulsive}}$$

Gravity

$$\vec{f}_g^{\,v} = [0 \;\; 0 \;\; mg]$$

$$\Rightarrow \vec{f}_g^{\,b} = R_v^b(\theta)\,\vec{f}_g^{\,b}$$

Aerodynamic Forces and Moments



Most aircraft have ailerons, elevators, rudders, flaps, etc. but they can always be expressed as something like the standard aileron-elevator-rudder setup

Pos deflection based on pos torque dir using RHR

$$F_{lift} = \tfrac{1}{2}\rho V_a^2 S_{ref} C_L(\alpha, q, \delta_e)$$
$$F_{drag} = \tfrac{1}{2}\rho V_a^2 S_{ref} C_D(\alpha, q, \delta_e)$$
$$m = \tfrac{1}{2}\rho V_a^2 S_{ref} c\, C_m(\alpha, q, \delta_e)$$

Longitudinal

$$F_{lift} = \tfrac{1}{2}\rho V_a^2 S_{ref}\left[C_{L_0} + \frac{\partial C_L}{\partial \alpha}\alpha + \frac{\partial C_L}{\partial q}q + \frac{\partial C_L}{\partial \delta_e}\delta_e\right] \Rightarrow \left(\text{not non-dimensional: } \frac{c}{2V_a}\right) = \tfrac{1}{2}\rho V_a^2 S_{ref}\left[C_{L_0} + \frac{\partial C_L}{\partial \alpha}\alpha + \frac{\partial C_L}{\partial\left(\frac{qc}{2V_a}\right)}\frac{c}{2V_a}q + \frac{\partial C_L}{\partial \delta_e}\delta_e\right]$$

$\frac{qc}{2V_a}$: nondimensionalize

$$F_{drag} = \tfrac{1}{2}\rho V_a^2 S_{ref}\left[C_{D_0} + \frac{\partial C_D}{\partial \alpha}\alpha + \frac{\partial C_D}{\partial\left(\frac{qc}{2V_a}\right)}\frac{c}{2V_a}q + \frac{\partial C_D}{\partial \delta_e}\delta_e\right]$$
$$m = \tfrac{1}{2}\rho V_a^2 S_{ref} c\left[C_{m_0} + \frac{\partial C_m}{\partial \alpha}\alpha + \frac{\partial C_m}{\partial\left(\frac{qc}{2V_a}\right)}\frac{c}{2V_a}q + \frac{\partial C_m}{\partial \delta_e}\delta_e\right]$$

Nothing accounts for stall, but a stall model (flat plate) can be spliced in ($C_{L_0} + C_{L_\alpha}\alpha = C_L(\alpha)$)

$C_{L,\,\text{flat plate}} = 2\,\text{sign}(\alpha)\sin^2\alpha\cos\alpha$ | Blending: $\sigma(\alpha) = \frac{1 + e^{-M(\alpha-\alpha_0)} + e^{M(\alpha+\alpha_0)}}{(1+e^{-M(\alpha-\alpha_0)})(1+e^{M(\alpha+\alpha_0)})}$ | α_0 = stall transition angle; M = transition rate

$$\Rightarrow C_L(\alpha) = (1-\sigma(\alpha))\left[C_{L_0} + \frac{\partial C_L}{\partial \alpha}\alpha\right] + \sigma(\alpha)\left[2\,\text{sign}(\alpha)\sin^2(\alpha)\cos(\alpha)\right]$$

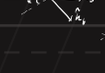
$$\Rightarrow C_D(\alpha) = (1-\sigma(\alpha))\left[C_{D_0} + \frac{\left(C_{L_0} + \frac{\partial C_L}{\partial \alpha}\alpha\right)^2}{\pi A R e}\right] + \sigma(\alpha)\cdot 2\sin^2(\alpha)\,|\sin(\alpha)|$$

Filling in the other half of the flat plate ($(+)\cos\alpha = L$ ∴ $(+)|\sin\alpha| = D$ (always opposite))

$$\Rightarrow C_m(\alpha) = C_{m_0} + \frac{\partial C_m}{\partial \alpha}\alpha$$ | simpler linear model because its hard to model

For small aircraft: $\frac{\partial C_L}{\partial \alpha} = \frac{\pi A R}{1 + \sqrt{1 + \left(\frac{AR}{2}\right)^2}}$

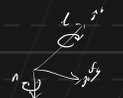
Conversion to body axes is a rotation by α



$$\Rightarrow f_x = L\sin\alpha - D\cos\alpha$$
$$f_z = -L\cos\alpha - D\sin\alpha$$
$$\Rightarrow \begin{bmatrix} f_x \\ f_z \end{bmatrix} = \begin{bmatrix} \sin\alpha & -\cos\alpha \\ -\cos\alpha & -\sin\alpha \end{bmatrix}\begin{bmatrix} D \\ L \end{bmatrix}$$

Lateral (Similarities to Longitudinal: L, m, n for moments and subscripts classic)

$$f_y = \tfrac{1}{2}\rho V_a^2 S_{ref} C_Y(\beta, p, r, \delta_a, \delta_r)$$
$$l = \tfrac{1}{2}\rho V_a^2 S_{ref} b\, C_l(\beta, p, r, \delta_a, \delta_r)$$
$$n = \tfrac{1}{2}\rho V_a^2 S_{ref} b\, C_n(\beta, p, r, \delta_a, \delta_r)$$

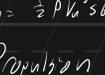


Same approach as before

$$f_y = \tfrac{1}{2}\rho V_a^2 S\left[C_{Y_0} + \frac{\partial C_Y}{\partial \beta}\beta + \frac{\partial C_Y}{\partial\left(\frac{pb}{2V_a}\right)}\frac{b}{2V_a}p + \frac{\partial C_Y}{\partial\left(\frac{rb}{2V_a}\right)}\frac{b}{2V_a}r + \frac{\partial C_Y}{\partial \delta_a}\delta_a + \frac{\partial C_Y}{\partial \delta_r}\delta_r\right]$$
$$l = \tfrac{1}{2}\rho V_a^2 S b\left[C_{l_0} + \frac{\partial C_l}{\partial \beta}\beta + \frac{\partial C_l}{\partial\left(\frac{pb}{2V_a}\right)}\frac{b}{2V_a}p + \frac{\partial C_l}{\partial\left(\frac{rb}{2V_a}\right)}\frac{b}{2V_a}r + \frac{\partial C_l}{\partial \delta_a}\delta_a + \frac{\partial C_l}{\partial \delta_r}\delta_r\right]$$
$$n = \tfrac{1}{2}\rho V_a^2 S b\left[C_{n_0} + \frac{\partial C_n}{\partial \beta}\beta + \frac{\partial C_n}{\partial\left(\frac{pb}{2V_a}\right)}\frac{b}{2V_a}p + \frac{\partial C_n}{\partial\left(\frac{rb}{2V_a}\right)}\frac{b}{2V_a}r + \frac{\partial C_n}{\partial \delta_a}\delta_a + \frac{\partial C_n}{\partial \delta_r}\delta_r\right]$$

already in body frame

Propulsion

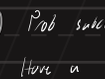


P_{thrust} ⊗ = Plus/minus

linear relationship between PWM command and V_{exit}^2 ⇒ $V_{exit} = k_{motor}\delta_t$ thrust itself ⇒ $F_{th} = S_{prop}C_{prop}(P_{thrust} - P_{ambient})$ ← Bernoulli's $= \tfrac{1}{2}\rho S_{prop}C_{prop}\left[(k_{motor}\delta_t)^2 - V_a^2\right]$

$$\Rightarrow \vec{f}_p = \tfrac{1}{2}\rho S_{prop}C_{prop}\begin{pmatrix}(k_{motor}\delta_t)^2 - V_a^2 \\ 0 \\ 0\end{pmatrix}$$

We assume ρ VCG is $\hat{x}_b$, passes through centroid, so no moments produced ⇒ $\vec{\tau}_p = \begin{bmatrix} -k_{\tau_p}(k_m\delta_t)^2 \\ 0 \\ 0 \end{bmatrix}$



f_p produces a torque, but has much denser on α, and blade geometry. $\Rightarrow \tau_p = -k_{\tau_p}(k_m\delta_t)^2$ for each motor

$F \propto \rho A\, V^2 \propto (k_m\delta_t)^2$ Assume to be much weaker than f_p and constant w.r.t. α

Atmospheric Disturbances

$$\underbrace{\vec{V}_g}_{\text{ground}} = \underbrace{\vec{V}_a}_{\text{air-mass}} + \underbrace{\vec{V}_w}_{\text{wind}}, \qquad \vec{V}_w = \underbrace{\vec{V}_{w_s}}_{\text{steady}} + \underbrace{\vec{V}_{w_g}}_{\text{stochastic gust}}$$

$\vec{V}_{w_s}$ is usually in inertial frame, $\vec{V}_{w_s}^{\,i} = \begin{bmatrix} u_{w_s} \\ v_{w_s} \\ w_{w_s} \end{bmatrix}$

$\vec{V}_{w_g}$ is usually in body frame, $\vec{V}_{w_g}^{\,b} = \begin{bmatrix} u_{w_g} \\ v_{w_g} \\ w_{w_g} \end{bmatrix}$

Stochastic Model (Dryden):

$$H_u(s) = \sigma_u\sqrt{\frac{2L_u}{\pi}}\frac{1}{s + \frac{V_a}{L_u}}$$
$$H_v(s) = \sigma_v\sqrt{\frac{3L_v}{\pi}}\frac{\left(s + \frac{V_a}{\sqrt{3}L_v}\right)}{\left(s + \frac{V_a}{L_v}\right)^2}$$
$$H_w(s) = \sigma_w\sqrt{\frac{3L_w}{\pi}}\frac{\left(s + \frac{V_a}{\sqrt{3}L_w}\right)}{\left(s + \frac{V_a}{L_w}\right)^2}$$

Transfer Functions

σ_i – intensity (how intense from axes)

L_i – spatial wavelengths

Two forms: (η = unit-intensity white noise, $n_k \sim N(0,1)$

$$H(s) = \frac{k}{s+a} \Rightarrow \dot{x} = -ax + \eta$$
$$u = Kx$$

$$H(s) = \frac{k(s+b)}{(s+a)^2}$$
$$(s+a)^2 = s^2 + 2as + a^2$$
$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -a^2 & -2a \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}\eta$$
$$V_g/V_a = K\cdot(bx_1 + x_2)$$

$$\Rightarrow \dot{\vec{x}} = \underline{A}\vec{x} + \underline{B}\eta \Rightarrow x_{n+1} = \vec{x}_n + \underline{A}\vec{x}_n\Delta t + \underline{B}\sqrt{\Delta t}\,n_k$$

random noise is scaled by $\sqrt{\Delta t}$ not Δt

Basically get n_k, advance x_i, get u_g, v_g, w_g, then in the DCM steps use $\vec{w}_b = \underline{R}(\epsilon)\cdot\vec{v}_{steady}^{\,i} + [u_g, v_g, w_g]^T$, and $\vec{V}_a = \vec{V}_b - \vec{w}_b$

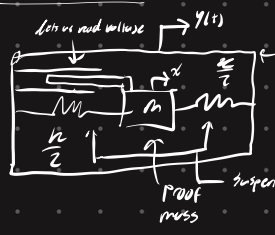
1.) Probably subclass the Body struct to implement different aircraft (config struct set by factory functions, and a trimmin method to be overriden for each type)

2.) Have a whole file of force models which accept the params, and we override the trimmin with those

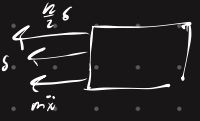
3.) Maybe reorganize into a "World", "Air System", "Body System"

Sensor Modelling

Accelerometers



$\delta = x - y(t)$



$\Rightarrow m\ddot{x} + k\delta = 0$

$\Rightarrow m\ddot{x} + kx = ky(t)$

$\mathcal{L}\{\cdots\}$

$\Rightarrow ms^2 X + kX = xY$

$H(s) = \frac{1}{\frac{m}{k}s^2+1} = \frac{X}{Y}$

$\Rightarrow \frac{s^2}{s^2} H(s) = \frac{A_x(s)}{A_y(s)} = \frac{1}{\frac{m}{k}s^2+1}$

and $\left|\frac{A_x}{A_y}\right|_{s=j\omega} = \frac{1}{1-\omega^2\frac{m}{k}} \approx 1$ for $\sqrt{\frac{k}{m}}$

(ie below resonant freq $\sqrt{\frac{k}{m}}$, δ is a direct measure of accel)

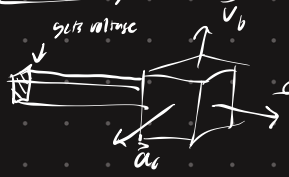
Gravity acts on both case and mass

$\therefore \vec{a}_{meas} = \frac{\vec{F} - m\vec{g}}{m} = \vec{a} - \vec{g}$

Model for actual voltage reading:

$\underbrace{v_{accel}}_{\text{volts}} = \underbrace{k_{accel}}_{\text{gain}}\underbrace{A}_{\text{accel in m/s}^2} + \underbrace{\beta_{accel}}_{\text{bias}} + \underbrace{\eta'_{accel}}_{\text{noise}}$, $\eta_{accel} \sim N(0, \sigma^2_{accel})$

Rate Gyros:



Coriolis accel:

$\vec{a}_c = 2\vec{\Omega} \times \vec{v}_b$

A proof mass is oscillated at resonance giving:

$v = \underbrace{A\omega}_{\text{amp}} \sin(\omega t)$, sensitive axis perp. to vibration

$\Rightarrow |\vec{\Omega} \times \vec{v}| = \Omega |v| \sin 90 = \Omega |v|$

$\Rightarrow V_{gyro} = 2k_c |\vec{\Omega} \times \vec{v}|$ (voltage)

$= 2k_c \Omega |A\omega \sin \omega t|$

$= 2k_c A\omega\Omega$

$= k_c \Omega$] angular vel about sensitive axis.

$\underbrace{v_{gyro}}_{\text{volts}} = \underbrace{k_{gyro}}_{\text{rad to volts}}\Omega + \underbrace{\beta_{gyro}}_{\text{bias}} + \underbrace{\eta'_{gyro}}_{\text{noise}}$ (converted to numerical by FC)

and having 3 aligned to body axes gives

$\underbrace{\vec{y}_{gyro}}_{\text{angular rate measurements}} = \vec{\omega}_{b/i} + \underbrace{\vec{\beta}_{gyro}}_{\text{bias}} + \underbrace{\vec{\eta}_{gyro}}_{\text{noise}}$, $\vec{\eta}_{gyro} \sim N(0, \vec{\sigma}^2_{gyro})$

Airspeed:

Pitot-tube $\Rightarrow P_{total} - P_{static} = \frac{1}{2}\rho V_a^2$

$\Rightarrow y_{diff,pas} = \frac{1}{2}\rho V_a^2 + p_{diff_p} + \eta_{diff,p}$, $\eta_{diff,p} \sim N(0, \sigma^2_{diff,p})$

Thus error at low airspeeds ($\propto V_a^2$), sampled at autopilot loop rate

Compass:

Tilt compensation:

$\vec{m}_b^{vl} = \underbrace{R_b^{vl}(\phi,\theta)}_{\text{body aligned magnetic measurements}}\vec{m}_b^{b}$

$\psi = \underbrace{\delta}_{\text{declination, got from error model}} + \psi_m$

$\psi_m \rightarrow -\text{atan2}(m_y^{vl}, m_x^{vl}) = \psi - \delta$ (magnetometers not aligned so this gives a heading component magnetometers will read)

Most don't internally, simply a simple model of:

$y_{mag} = \psi + p_{mag} + \eta_{mag}$, $\eta_{mag} \sim N(0, \sigma^2_{mag})$

(converted to $[-\pi, \pi]$)

with single rate T_s

$\alpha = e^{-k_r T_s}$ $v[n], y[n]$ input

$Var(v[n+1]) = Var(v[n]) + \sigma_\eta^2$

Steady $Var(v[n+1]) = Var(v[n])$

$\Rightarrow \sigma_v = \frac{\sigma_\eta}{\sqrt{1-e^{-2k_r T_s}}}$

Conversion to body axes (ie accounting for Coriolis)

$\vec{a} = \frac{d\vec{v}}{dt_i} = \frac{d\vec{v}}{dt_b} + \vec{\omega}_{b/i} \times \vec{V}$, $\frac{d\vec{v}}{dt_b} = \begin{bmatrix}\dot{u}\\ \dot{v}\\ \dot{w}\end{bmatrix}$, $\vec{\omega}_{b/i} = \begin{bmatrix}p\\ q\\ r\end{bmatrix}$

$\Rightarrow \vec{a}_{meas} = \vec{a} - R_v^b \begin{pmatrix}0\\0\\g\end{pmatrix}$

$\Rightarrow \begin{bmatrix}a_x\\ a_y\\ a_z\end{bmatrix} = \begin{bmatrix}\dot{u} + qw - rv + g\sin\theta\\ \dot{v} + ru - pw - g\cos\theta\sin\phi\\ \dot{w} + pv - qu - g\cos\theta\cos\phi\end{bmatrix}$

The output voltage can become an accel. reading in the autopilot microcontroller through an analog-digital converter at a sample rate T_s. Signals in the autopilot can then be modelled as:

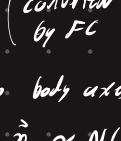
$\vec{y}_{accel} = \vec{a}_{meas} + \underbrace{\vec{\eta}_{accel}}_{\text{zero mean gaussian noise w. } \sigma_a^2 \text{ variance}}$

We already have forces computed and

$\Rightarrow \begin{bmatrix}\dot{u}\\ \dot{v}\\ \dot{w}\end{bmatrix} = \begin{bmatrix}rv - qw\\ pw - ru\\ qu - pv\end{bmatrix} + \frac{1}{m}\begin{bmatrix}f_x\\ f_y\\ f_z\end{bmatrix} \Rightarrow$ Coriolis cancels and each component collapses to $\frac{1}{m}\vec{F}$

$\Rightarrow \vec{y}_{accel} = \frac{\vec{F} - F_g(m,\vec{\theta})}{m} + \underbrace{\vec{\beta}_{accel}}_{\text{assumed calibrated away}} + \vec{\eta}_{accel}$, $\eta \sim N(0, \sigma^2_{accel})$

Altitude:



$P(z)A - P(z+dz)A + \rho g A\,dz = 0$

$\frac{dP}{dz} = \rho g \Rightarrow P_2 - P_1 = \rho g (z_2 - z_1)$

$\Rightarrow P - P_{ground} = -\rho g(h - h_{ground}) = -\rho g h_{AGL}$

$\rho = \frac{dM}{dV}$, $T = T_0 + \underbrace{L_0}_{\text{lapse}} h$ (below 11,000 m, otherwise just using ISA model)

$\frac{dP}{P/h} = -\frac{g_0}{R}\frac{\rho}{T_0 + L_0 h} \Rightarrow \int_{P_0}^{P_{ab}} \ln\frac{P}{P_0} = -\frac{g_0 M}{R L_0}\ln\frac{T_0 + L_0 h}{T_0}$

$\Rightarrow P = P_0\left[\frac{T_0}{T_0 + L_0 h_{abs}}\right]^{\frac{g_0 M}{R L_0}}$

Linearising (they might use ISA):

$P = P_0 - \underbrace{P_0 g h}_{-\Delta P}$ Sampled at autopilot loop rate

$\Rightarrow y_{abs,pas} = \rho g h_{AGL} + p_{abs,pas} + \eta_{abs,pas}$, $\eta \sim N(0, \sigma^2)$

GPS:

Steady error has many components (T 7.1), adding them as an RMS gives a user equivalent range error (UERE) and a second dilution of precision (DOP), which are multiplied.

$E_{n-e,rms} = HDOP \times UERE_{rms} = 6.6\,m$, $\sigma_n^2 + \sigma_e^2 = 2\sigma_n^2 \Rightarrow \sigma_n = 4.7\,m$

$E_{h,rms} = VDOP \times UERE_{rms} = 9.2\,m$, $\sigma_h = 3.03\,m$

Transient err modelled with Markov process

$v[n+1] = e^{-k_{GPS}T_s} v[n] + \eta_{GPS}[n]$, $\eta \sim N(0, \sigma_\eta^2)$

For white accel, $1 - e^{-2k_r T_s} \approx 2k_r T_s$ so $\sigma_v \approx \frac{\sigma_\eta}{\sqrt{2k_r T_s}}$

and $y_{gps,n} = p_n + v_n$, $y_{gps,e} = p_e + v_e$, $y_{gps,h} = -p_d + v_h$

GPS Velocity Measurement

north/east ground vel → $v_g = \sqrt{v_n^2 + v_e^2}$, course → $\chi = \text{atan2}(v_e, v_n)$

err. prop. $\sigma_f^2 = \left(\frac{\partial f}{\partial v_n}\right)^2 \sigma_{v_n}^2 + \left(\frac{\partial f}{\partial v_e}\right)^2 \sigma_{v_e}^2$

$\frac{\partial v_g}{\partial v_n} = \frac{v_n}{v_g}$, $\frac{\partial v_g}{\partial v_e} = \frac{v_e}{v_g}$, $\sigma_{v_g}^2 = \frac{v_n^2 \sigma_{v_n}^2 + v_e^2 \sigma_{v_e}^2}{v_n^2 + v_e^2}$

and $\frac{d}{dx}(\tan^{-1}(u)) = \frac{\frac{du}{dx}}{1+u^2}$, $u = \frac{v_e}{v_n}$

$\Rightarrow \frac{\partial \chi}{\partial v_n} = \frac{-v_e/v_n^2}{1 + v_e^2/v_n^2} = -\frac{v_e}{v_g^2}$, $\frac{\partial \chi}{\partial v_e} = \frac{1/v_n}{1 + v_e^2/v_n^2} = \frac{v_n}{v_g^2}$

$\sigma_\chi^2 = \frac{v_n^2 \sigma_{v_n}^2 + v_n^2 \sigma_{v_e}^2}{(v_n^2 + v_e^2)^2}$, and with $\sigma_{v_n} = \sigma_{v_e} = \sigma_v$

numerators both factor to $\sigma_v^2 v_g^2$

$\Rightarrow \sigma_{v_g} = \sigma_v$, $\sigma_\chi = \frac{\sigma_v}{v_g}$

Sim already has inertial readings for $\vec{v}$ ($\vec{v}^b = \underline{R}_b^v \vec{v}^v$)

$\Rightarrow y_{gps,v_g} = \sqrt{v_n^2 + v_e^2} + \eta_v$, $y_{gps,\chi} = \text{atan2}(v_e, v_n) + \eta_\chi$

$\eta_{v,i} \sim N(0, \sigma_v^2)$, $\eta_\chi \sim N(0, \sigma_\chi^2)$

$\sigma_v^2 = \sigma_{v_n}^2 = \sigma_{v_e}^2$

GPS usually sampled every 1s

Basically just another params + functions / decay sampling sensors after updating the body. Each sensor should just have a sensor member containing the readings.

Control (just the basic PID)

$\underbrace{u(t)}_{\text{ctrl. out}} = k_p \underbrace{e(t)}_{\text{err. signal}} + k_i \int_0^t e(\tau)\,d\tau + k_d \frac{de(t)}{dt} + \underbrace{ff}_{\text{any feedforward signal}}$

also have some saturation state

Integrator:

Trapezoidal:

$y[n] = y[n-1] + \frac{dt}{2}(u[n] + u[n-1])$

Band Limited Differentiator:

$\frac{s}{\tau s + 1}$ | $s = \frac{2}{dt}(z-1)/(z+1)$

$\Rightarrow y[n] = \frac{2\tau - dt}{2\tau + dt} y[n-1] + \frac{2}{2\tau + dt}(u[n] - u[n-1])$

Lowpass Filter

$1/s$, some substitution

$y[n] = \frac{2\tau - dt}{2\tau + dt} y[n-1] + \frac{dt}{2}(2\tau + dt)(u[n] + u[n-1])$

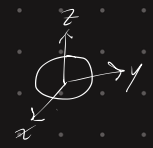
Simple PID implementation with the same params struct structure

Should probably store visible state separately (see if maybe a $\underline{C}$ matrix is better).

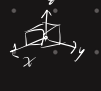
Look at the more complicated setup later, should be sufficient for doing a simple satellite model

Satellites

New inertial frame i (ECI)



Body Frame



Satellite body (can just reuse what I have)

$\vec{v}^i = R\vec{v}^b$ and $\vec{\omega}$ still in body frame

R from attitude quat

Earth is wider at equator (J_2)

$\vec{a} = -\nabla U$, (only using latitude oblateness term)

$\sin(\text{lat}) = \frac{z}{r}$, $U = -\frac{\mu}{r}\left[1 - J_2\left(\frac{R_E}{r}\right)^2 P_2\left(\frac{z}{r}\right)\right]$, $P_2(x) = \frac{3x^2-1}{2}$ } this is cool, i've never seen this

$\Rightarrow U = -\frac{\mu}{r} + \frac{\mu J_2 R_E^2}{r^3}\left(\frac{3z^2}{r^2} - \frac{1}{2}\right)$

$\Rightarrow \frac{\partial u}{\partial \vec{r}} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial \vec{r}}, \frac{\partial r}{\partial \vec{r}} = \ldots, \frac{\partial r}{\partial \vec{r}} = \frac{\vec{r}}{r}\ (r = \ldots)$

$\frac{\partial}{\partial x}\left(\frac{1}{r}\right) = -\frac{x}{r^3}, \frac{\partial}{\partial x}\left(\frac{1}{r^3}\right) = \frac{3x}{r^5}, \frac{\partial}{\partial x}\left(\frac{R_E^2}{r^5}\right) = \frac{5xz^2}{r^7}$ | y identically

$\frac{\partial}{\partial z}\left(\frac{1}{r}\right) = -\frac{z}{r^3}, \frac{\partial}{\partial z}\left(\frac{z^2}{r^5}\right) = \frac{2z}{r^5} - \frac{5z^3}{r^7}$

$\frac{\partial u}{\partial x} = \frac{\mu x}{r^3} + \frac{\mu J_2 R_E^2}{2}\left(\frac{-15x^2}{r^7} + \frac{3x}{r^5}\right) = \frac{\mu x}{r^3} + \frac{3\mu J_2 R_E^2 x}{2r^5}\left(1 - \frac{5z^2}{r^2}\right)$

$\frac{\partial u}{\partial z} = \frac{\mu z}{r^3} + \frac{\mu J_2 R_E^2}{2}\left(\frac{6z}{r^5} - \frac{15z^3}{r^7} + \frac{3z}{r^5}\right) = \frac{\mu z}{r^3} + \frac{3\mu J_2 R_E^2 z}{2r^5}\left(3 - \frac{5z^2}{r^2}\right)$

$\vec{a} = -\nabla u$, $\frac{\mu}{r^3}$ is common factor

$a_x = -\frac{\mu x}{r^3}\left[1 + \frac{3}{2}J_2\left(\frac{R_E}{r}\right)^2\left(1 - \frac{5z^2}{r^2}\right)\right]$

$a_y = -\frac{\mu y}{r^3}\left[1 + \frac{3}{2}J_2\left(\frac{R_E}{r}\right)^2\left(1 - \frac{5z^2}{r^2}\right)\right]$

$a_z = -\frac{\mu z}{r^3}\left[1 + \frac{3}{2}J_2\left(\frac{R_E}{r}\right)^2\left(3 - \frac{5z^2}{r^2}\right)\right]$

$J_2 = 1.0826\times10^{-3}$, $R_E = 6.378\times10^6$

J_2 Potential doesn't change with time in ECI as is symmetric about spin axis

Basic Forces

$m\ddot{\vec{r}}_s = -\frac{GMm}{\|\vec{r}\|^3}\vec{r}$, $\vec{r} = \vec{r}_s - \vec{r}_c$

$\|\vec{r}\| = r$ (choose to write it out)

$M\ddot{\vec{r}}_c = \frac{GMm}{\|\vec{r}\|^3}\vec{r}$

$\Rightarrow \ddot{\vec{r}} = \ddot{\vec{r}}_s - \ddot{\vec{r}}_c = -\frac{G(M+m)}{\|\vec{r}\|^3}\vec{r}$

With $m \ll M$, letting $\mu = GM$

$\ddot{\vec{r}} = -\frac{\mu}{\|\vec{r}\|^3}\vec{r}$, $\mu = 3.986\times10^{14}\ m^3/s^2$

and rotating external forces $\vec{F}^b$ into inertial frame

$\Rightarrow \ddot{\vec{r}} = -\frac{\mu}{\|\vec{r}\|^3}\vec{r} + \frac{1}{m}\underline{R}\vec{F}^b$

Conserved quantities are

$\mathcal{E} = \frac{1}{2}v^2 - \frac{\mu}{r}$

i.e. $\frac{d}{dt}\left(\frac{1}{2}\vec{v}\cdot\vec{v}\right) = \vec{v}\cdot\dot{\vec{v}} = -\frac{\mu}{r^3}\vec{v}\cdot\vec{r}$

$\frac{d}{dt}\left(-\frac{\mu}{r}\right) = \frac{\mu}{r^2}\dot{r}$, $r\cdot\dot{r} = \frac{1}{2}\frac{d}{dt}(\vec{r}\cdot\vec{r}) = \vec{r}\cdot\vec{v}$

$\Rightarrow \frac{\mu}{r^2}\dot{r} = \frac{\mu}{r^3}\vec{r}\cdot\vec{v}$ ← cancels

$\Rightarrow \dot{\mathcal{E}} = 0$

$\vec{h} = \vec{v}\times\vec{v} + \vec{r}\times\left(-\frac{\mu}{r^3}\vec{r}\right) = 0$ ✓

Initial conds is circular orbit

$\Rightarrow \frac{\mu}{r^2} = \frac{v^2}{r}$ } $m\frac{v^2}{r} = F$ (centripetal)

$\Rightarrow v_c = \sqrt{\frac{\mu}{r}}$, $T = \frac{2\pi r}{v_c} = 2\pi\sqrt{\frac{r^3}{\mu}}$

Letting $\vec{r}_0 = \begin{bmatrix} r \\ 0 \\ 0 \end{bmatrix}$, $\vec{v}_0 = v_c\begin{bmatrix} 0 \\ \cos\theta \\ \sin\theta \end{bmatrix}$

Torque

Euler: $\vec{\tau} = \left.\frac{d\vec{L}}{dt}\right|_i$ (transport $\left.\frac{d}{dt}\right|_i[\cdot] = \left.\frac{d}{dt}\right|_b[\cdot] + \omega\times[\cdot]$)

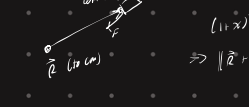
$\Rightarrow \vec{\tau} = \left.\frac{d}{dt}[\underline{I}\vec{\omega}]\right|_i = \left.\frac{d}{dt}[\underline{I}\vec{\omega}]\right|_b + \vec{\omega}\times(\underline{I}\vec{\omega})$ | assuming $\underline{I}$ constant in body axes

$\Rightarrow \underline{I}\dot{\vec{\omega}} + \vec{\omega}\times(\underline{I}\vec{\omega}) = \vec{\tau}$ (already have this)

and $\dot{\vec{\omega}} = \underline{I}^{-1}(\vec{\tau} - \vec{\omega}\times(\underline{I}\vec{\omega}))$

Gravity Gradient Torque

Gravity acts slightly differently at separate ends of the aircraft



$d\vec{F} = \frac{-\mu(\vec{R}+\vec{\delta})}{\|\vec{R}+\vec{\delta}\|^3}dm$

$(1+x)^{-2} \approx 1 - \frac{2}{2}x$, $\|\vec{\delta}\|^2 \approx 0$

$\Rightarrow \|\vec{R}+\vec{\delta}\|^{-3} = R^{-3}\left(1 + \frac{2\vec{R}\cdot\vec{\delta}}{R^2} + \frac{\delta^2}{R^2}\right)^{-\frac{3}{2}}$

$\approx R^{-3}\left(1 - \frac{3\vec{R}\cdot\vec{\delta}}{R^2}\right)$

$\vec{\tau} = \int\vec{\delta}\times d\vec{F} = -\frac{\mu}{R^3}\int\vec{\delta}\times(\vec{R}+\vec{\delta})\left(1 - \frac{3\vec{R}\cdot\vec{\delta}}{R^2}\right)dm$

$\vec{\delta}\times\vec{\delta} = 0$

$\int\vec{\delta}\,dm = 0$ (its about com)

$\Rightarrow \vec{\tau} = \frac{3\mu}{R^5}\int(\vec{R}\cdot\vec{\delta})(\vec{\delta}\times\vec{R})dm = -\frac{3\mu}{R^5}\vec{R}\times\left(\int\vec{\delta}\vec{\delta}^T dm\right)\vec{R}$

$\vec{L} = \int\underbrace{\vec{\delta}}_{\text{moves}}dm\times\underbrace{(\vec{\omega}\times\vec{\delta})}_{\text{velocity relative com}}$

$\vec{\delta}\times(\vec{\omega}\times\vec{\delta}) = \vec{\omega}(\vec{\delta}\cdot\vec{\delta}) - \vec{\delta}(\vec{\delta}\cdot\vec{\omega})$ (identity)

$\Rightarrow \|\vec{\delta}\|^2 I - \vec{\delta}\vec{\delta}^T)\vec{\omega}$

and $\vec{L} = \underline{I}\vec{\omega} \Rightarrow \underline{I} = \int(\|\vec{\delta}\|^2 I - \vec{\delta}\vec{\delta}^T)dm$

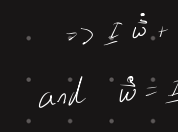
⇒ I honestly couldn't finish this, don't understand remainder

$\vec{\tau} = \frac{3\mu}{\|\vec{r}\|^5}\vec{r}^b\times\underline{I}\vec{r}^b$, $\vec{r}^b = \underline{R}^T\vec{r}$ | from Claude

Reaction Wheels

$\underline{I} = \underbrace{\underline{I}_S}_{\text{Craft}} + \underbrace{\underline{I}_w}_{\text{Wheels}}$

Three reaction wheels



using $\hat{a}$ as spin (I instead of I_w)

$I_{wheel} = \begin{bmatrix} I_T & 0 & 0 \\ 0 & I_T & 0 \\ 0 & 0 & I_w \end{bmatrix} = \underbrace{I_T\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{I - \hat{a}\hat{a}^T} + \underbrace{I_w\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\hat{a}\hat{a}^T}$ } i.e. $(\hat{a}\hat{a}^T)\hat{a} = \hat{a}(\hat{a}\cdot\hat{a})$

rotating to body $\Rightarrow \underline{R}I_w\underline{R}^T$ (taking $\hat{a}$ to $\hat{a}$).

$R(\hat{a}\hat{a}^T)R^T = (R\hat{a})(R\hat{a})^T = \hat{a}\hat{a}^T$, $RER^T = E$

$\Rightarrow I_{wheel} = I_T\underbrace{(I - \hat{a}\hat{a}^T)}_{\text{across axis, perp}} + \underbrace{I_w\hat{a}\hat{a}^T}_{\text{along axis}}$ | $I_{wheel}\hat{a}_0 = I_w\hat{a}$

Wheel spins about its axis and body also spins

$\Rightarrow \Omega_n\hat{a}_n = \vec{\omega}_n - \vec{\omega}$

$\Rightarrow L_n = I_{wheel}(\vec{\omega} + \Omega_n\hat{a}_n) = \underbrace{I_{wheel}\vec{\omega}}_{\text{any momentum due to spin about its axis}} + \underbrace{I_w\Omega_n\hat{a}_n}_{\text{any momentum from it spinning itself}}$

$\Rightarrow \vec{L} = \underline{I}\vec{\omega} + \sum_n I_w\Omega_n\hat{a}_n$

$\vec{\tau}_{ext} = \underline{I}\dot{\vec{\omega}} + \sum_n I_w\dot{\Omega}_n\hat{a}_n + \vec{\omega}\times(\underline{I}\vec{\omega})$ ($\hat{a}_n$ is fixed in the body)

$\vec{L}_n = I_T\vec{\omega}_\perp + \vec{I}_w(\hat{a}\cdot\vec{\omega} + \Omega_n)\hat{a}$

$\vec{L}_\perp = \vec{L} - (\hat{a}\cdot\vec{L})\hat{a}$

$\vec{\omega}\times\hat{a}$ is perp to both $\vec{\omega}$ and $\hat{a}$, so its perp to $\vec{\omega}_\perp$ and $\hat{a} \Rightarrow \dot{L}_n(\vec{\omega}\times\hat{a}) = 0$

$\left.\frac{d\hat{a}}{dt}\right|_i = \vec{\omega}\times\hat{a}$

τ_{motor} axis lines with body $\Rightarrow \frac{d}{dt}(\hat{a}\cdot\vec{L}_n) = \hat{a}\cdot\left.\frac{d\vec{L}_n}{dt}\right|_b + \underbrace{\dot{L}_n\cdot(\vec{\omega}\times\hat{a})}_{=0}$

$\Rightarrow \tau_{motor} = I_w(\dot{\Omega}_n\cdot\dot{\omega} + \dot{\Omega}_n)$

Directly from Claude

$A = [\hat{a}_1 \ldots \hat{a}_n]$, so many $m_n = \hat{a}_n\cdot\vec{L}_n = I_w(\dot{\Omega}_n\cdot\vec{\omega} + \dot{\Omega}_n)$, $\dot{m}_n = \tau_{motor,n}$

so $\dot{m} = A^T\vec{\tau}_{ext}$, $\dot{m} = A^T\vec{\tau}_{motor}$

$\dot{\vec{\omega}} = \underline{I}_S^{-1}(\vec{\tau}_{ext} - A\vec{\tau}_{motor} - \vec{\omega}\times(I_S\vec{\omega} + \vec{h}_w))$

$I_S = \underline{I} - \sum_n I_w\hat{a}_n\hat{a}_n^T$

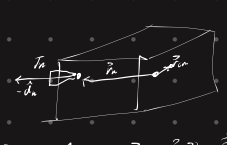
For my case $\hat{a}_w = \hat{x},\hat{y},\hat{z}$

$I_S = \underline{I} - I_w\underline{I}$

Honestly also can't go further here, but the final result is

$\Rightarrow$

Thrusters



$\vec{F}_n = \underbrace{T_n\hat{a}_n}_{\text{Thrust dir}}$, $\vec{\tau}_m = (\vec{r}_0 - \vec{r}_{cm})\times\vec{F}_n$

$\vec{F}^b_{thr} = \sum_n \vec{F}_n$ $\tau_{thr} = \vec{\tau}\,\vec{r}_n$

Mass:

$T\,dt = v_e\,dm$

$\Rightarrow T = \dot{m}v_e$

$I_{sp} = \frac{v_e}{g_0}$ (gravity at surface)

} $\dot{m} = -\sum_n \frac{T_n}{I_{sp,n}g_0}$

Most of the gravity infrastructure should be able to handle mass.

- mass loss, reaction wheels, thrusters ✓

Unfortunately I could only understand some of the derivations (I couldn't for reaction wheels and the gravity gradient torque unfortunately) But final eqns should be easy to implement.

State Estimation

Low Pass?

$$Y(s) = \frac{a}{s+a}\mathcal{L}\{u(t)\}$$ (filter input)

Inverse gives:

$$\dot{y} = -ay(t) + au(t)$$

and $y(t+\bar{t}_s) = e^{-a\bar{t}_s}y(t) + a\int_0^{\bar{t}_s} e^{-a(\bar{t}_s - z)}u(z)\,dz$ ("convolution" over $\bar{t}_s$ with $g(t) = e^{-at}$)

assuming $u(t)$ const over $\bar{t}_s$ period

$$\Rightarrow y[n+1] = e^{-a\bar{t}_s}y[n] + a\int_0^{\bar{t}_s} e^{-a(\bar{t}_s - z)}dz\,u[n]$$ (I remember this from mechatronics)

$$= e^{-a\bar{t}_s}y[n] + (1 - e^{-a\bar{t}_s})u[n]$$

$$\Rightarrow \alpha_{LPF} = e^{-a\bar{t}_s}$$

$$\Rightarrow y[n+1] = \alpha_{LPF}y[n] + (1-\alpha_{LPF})u[n]$$

Apply to relevant sensors

Dynamic - Observer Theory

$$\dot{\underline{x}} = \underline{A}\underline{x} + \underline{B}\underline{u}$$

$$\underline{y} = \underline{C}\underline{x}$$

A continuous observer is given by (estimated val over $\dot{x}$):

$$\dot{\hat{x}} = \underbrace{A\hat{x} + Bu}_{\text{copy of model}} + \underbrace{\underline{L}(y - C\hat{x})}_{\text{correction due to sensor reading}}$$

using $\tilde{x} = x - \hat{x}$ (observation err) $\Rightarrow \dot{\tilde{x}} = (\underline{A} - \underline{L}\underline{C})\tilde{x}$ (actual sensor measurement) $= \hat{x}^-$ (decays to zero if eigenvalues in LHP)

$$\hat{x}^+ = \hat{x}^- + \underline{L}(\underbrace{y(t_n)}_{\text{measurement}} - C\hat{x}^-)$$

$$\Rightarrow \dot{\hat{x}} = \underline{f}(\hat{x}, \underline{u})$$

and $\hat{x}^+ = \hat{x}^- + \underline{L}(y(t_n) - h(\hat{x}^-))$ (Observer gain: how far off is our estimation based on how our sensors say)

} basically get estimated state, and using sensors and estimated state get a new more accurate estimated state

Kalman Filter

$\dot{\underline{x}} = \underline{A}x + Bu + \underline{\xi}$, ξ zero mean gaussian with covariance Q, process noise (and is system known).

$y[n] = Cx[n] + \eta[n]$ η zero mean gaussian with covariance R, sensor noise

From this:

$$\dot{\hat{x}} = A\hat{x} + Bu$$
$$\hat{x}^+ = \hat{x}^- + L(y(t_n) - C\hat{x}^-)$$

$\tilde{x} = x - \hat{x}$ and covariance of estimation error at time t

$P(t) = E\{\tilde{x}(t)\tilde{x}(t)^T\}$ (semi definite symmetric)

We want to choose $L(t)$ that minimizes eigenvalues of $P(t)$

$\Rightarrow tr(P) = \sum_i^n \lambda_i \Rightarrow$ Min $tr(P)$ minimizes error covariance

Between measurements

$$\dot{\tilde{x}} = \dot{x} - \dot{\hat{x}}$$ (how error is moving)

$$= \underbrace{Ax + Bu}_{\text{real}} + \xi - \underbrace{A\hat{x} - Bu}_{\text{estimated}}$$

$$= A\tilde{x} + \xi$$

Solving with init cond $\tilde{x}_0$

$$\tilde{x}(t) = e^{At}\tilde{x}_0 + \int_0^t e^{A(t-\tau)}\xi(\tau)\,d\tau$$ (old sensor noise matters less than newer)

$$\dot{P} = \frac{d}{dt}E\{\tilde{x}\tilde{x}^T\} = E\{\dot{\tilde{x}}\tilde{x}^T + \tilde{x}\dot{\tilde{x}}^T\}$$

$$= E\{A\tilde{x}\tilde{x}^T + \xi\tilde{x}^T + \tilde{x}\tilde{x}^TA^T + \tilde{x}\xi^T\}$$

$$= AP + PA^T + E\{\xi\tilde{x}^T\} + E\{\tilde{x}\xi^T\}$$

and $E\{\tilde{x}\xi^T\} = E\{e^{At}\tilde{x}_0\xi^T(t) + \int_0^t e^{A(t-\tau)}\xi(\tau)\xi(t)\,d\tau\}$

$$= \int_0^t e^{A(t-\tau)}Q\delta(t-\tau)\,d\tau$$ (honestly stating to self to look here, review later!)

$$= \frac{1}{2}Q$$

$$\Rightarrow \dot{\underline{P}} = \underline{A}\underline{P} + \underline{P}\underline{A}^T + \underline{Q}$$

At measurement:

$$\tilde{x}^+ = x - \hat{x}^+$$

$$= x - \hat{x}^- - L(Cx + \eta - C\hat{x}^-)$$

$$= \tilde{x}^- - LC\tilde{x}^- - L\eta$$

honestly the rest of this derivation goes over my head so I won't say I understand it.

I will implement the pseudo code from the textbook for EKF

From my understanding, we have a model (eg integrating gyro readings) which drifts and sensors which are slow, noisy or only see some of the state.

$\hat{x}$ is our current best estimate of the state, and P is how uncertain each part of $\hat{x}$ is and how those errors relate. (ie if position estimate is to high, velocity will probably also be too high)

So!

$\hat{x} \leftarrow f(\hat{x}, u)$ push the estimate through the dynamics

$P \leftarrow FPF^T + Q$, push the uncertainty through, and add Q, the process noise (so gyro noise, unmodeled forces, etc)

then!

$\tilde{y} = y - h(\hat{x})$, what the sensors say is true versus what was expected

and finally!

$\hat{x} \leftarrow \tilde{x} + K\tilde{y}$, so update estimate based on how we are wrong

P gets updated based on that estimate as well (multiplied by a factor), giving us a new system uncertainty

REPEAT!!!

Reaction Wheel Pointing Control

Plant

Splitting tot angular momentum about CM into body + wheel transverse inertia and wheel spin

$$\vec{L}^B = (\underline{I} - I_w \underline{A}\underline{A}^T)\vec{\omega} + \sum_k I_w(\vec{a}_k^T\vec{\omega} + \underbrace{s_k}_{\text{rel spin to body}})\vec{a}_k = \underbrace{\underline{I}_s\vec{\omega}}_{\text{body + wheel inertia}} + \underbrace{\underline{A}\vec{L}}_{\text{wheels spinning}} \Big\} \text{ tot ang mom}$$

$\underline{A} = [a_1 \dots]$ wheel axes in body, $\vec{L}$ = wheel ang mom, spin axis

$$\Rightarrow \dot{\vec{L}}^B + \vec{\omega}\times\vec{L}^B = \vec{\tau}_{ext}$$

and $\dot{\vec{L}} = \vec{\tau}_{mot}$ (no fric)

$$\Rightarrow \underline{I}_s\dot{\vec{\omega}} = \vec{\tau}_{ext} - \underline{A}\vec{\tau}_m - \vec{\omega}\times(\underline{I}_s\vec{\omega} + \underline{A}\vec{L}) \Big\} \text{ Plant}$$

The body feels $-\underline{A}\vec{\tau}_m$ so a body torque $\vec{u} = -A^{-1}\vec{\tau}_m$

and $\underline{I}_s\dot{\vec{\omega}} = \vec{u} + \vec{\tau}_{ext} - \vec{\omega}\times\vec{L}^B$ | Saturated so $\vec{\tau}_m \le \vec{\tau}_{max}\cdot\min\left(1, \frac{\tau_{max}}{\|\tau_m\|}\right)$

$\vec{L}^B = \underline{R}\vec{L}^B$, $\vec{L}^E = \underline{R}\vec{L}^B$

if starting at rest $\vec{L}_I = 0$ and with no ext torque $\dot{\vec{L}}^E = 0$ so $\vec{L}^E$ is const

$\Rightarrow \underline{A}\vec{L} = -\underline{I}_s\vec{\omega}$, $\underline{I}_s\dot{\vec{\omega}} = \vec{u}$ (i.e. $\vec{\omega}\times H_B = 0$) } singularities stable for inertial

From my Quat rate function earlier

$q = \frac{1}{2}q \otimes \begin{bmatrix}0\\ \vec{\omega}\end{bmatrix} \to \dot{e}_0 = -\frac{1}{2}\vec{\varepsilon}^T\vec{\omega}$, $\dot{\vec{\varepsilon}} = \frac{1}{2}(e_0\vec{\omega} + \vec{\varepsilon}\times\vec{\omega})$

Attitude Err:

(any rotation is a rotation ψ about an axis $\hat{n}$)

$$q_e = \underbrace{q_d^*}_{\text{desired}} \otimes \underbrace{q}_{\text{current}} = \begin{bmatrix}e_0\\ \vec{\varepsilon}_e\end{bmatrix} = \begin{bmatrix}\cos(\psi/2),\\ \sin(\psi/2)\hat{n}\end{bmatrix}$$ | $\hat{n}$ in body axes

q_e and $-q_e$ are the same attitude so we can flip to keep $\psi\in[0,\pi]$ (i.e. $e_0 \ge 0$)

then:

$$e = 2\,\mathrm{sgn}(e_0)\vec{\varepsilon}_e = 2\sin(\psi/2)\hat{n} \approx \delta\vec{\theta}$$ | each component is the resp. angle error about that body axis

and for small error $q_e \approx [1, \frac{1}{2}\delta\theta]$

For a fixed boresight $\dot{q}_e = q_d^* \otimes \dot{q}$ $\Rightarrow \dot{e}_{0,e} = -\frac{1}{2}\vec{\varepsilon}_e^T\vec{\omega}$, $\dot{\vec{\varepsilon}}_e = \frac{1}{2}(e_{0,e}\vec{\omega} + \vec{\varepsilon}_e\times\vec{\omega})$

and $\mathrm{sgn}(e_{0e})$ only changes after passing through 180° so const in between

$$\Rightarrow \dot{\hat{e}} = \mathrm{sgn}(e_{0e})(e_{0e}\vec{\omega} + \vec{\varepsilon}_e\vec{\omega}) \approx \vec{\omega}$$ | as $e_{0,e}\to 1$ and $\vec{\varepsilon}_e \to 0$ near the boresight

Linear Model and PID gains

$\Rightarrow$

$I_s\dot{\vec{\omega}} = \vec{u}$ and $\dot{e} \approx \vec{\omega} \Rightarrow I_s\ddot{e} \approx \dot{\vec{u}} + \tau_{ext}$

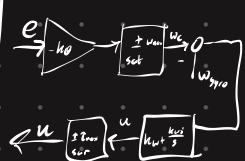
using just PD with rate from gyro $\vec{u} = -\underline{k}_p\vec{e} - \underline{k}_d\vec{\omega}$

and with $\underline{k}_p = \omega_n^2\underline{I}_s$, $\underline{k}_d = 2\zeta\omega_n\underline{I}_s$ $\Rightarrow \ddot{\vec{e}} + 2\zeta\omega_n\dot{\vec{e}} + \omega_n^2\vec{e} = \underline{I}_s^{-1}\vec{\tau}_{ext}$

Assuming off diag terms are small for I_s

$\Rightarrow k_p = I_{ss}\omega_n^2$, $k_d = 2\zeta\omega_n I_{ss}$, $e_{ss} = \frac{\tau_d}{k_p}$ } steady error under const torque τ_d

The controller



Outer loop turns attitude error into rate command, then the inner loop tracks the rate command with a torque:

$\omega_c = \mathrm{sat}_{\omega_{max}}(-k_0 e)$ commanded rate (outer)

$u = \mathrm{sat}_{\tau_{max}}\left(k_w(\omega_c - w) + k_{wi}\int(\omega_c - w)dt\right)$ Cmd (inner)

Inner:

$J\dot{\omega} = k_w(\omega_c - \omega)$ bandwidth

$\Rightarrow \omega_c = \frac{1}{(J/k_w)s+1}$ $\Rightarrow k_w = J_{ss}\omega_w$

Tuner:

$u = k_w(-k_0 e - w) = -\underbrace{k_w k_0}_{k_p} e - \underbrace{k_w}_{k_d} w$

$\omega_n = \sqrt{\omega_w\omega_0}$, $\zeta = \frac{1}{2}\sqrt{\frac{\omega_w}{\omega_0}}$ $\Leftrightarrow \frac{\omega_w}{\omega_0} = 4\zeta^2$

Pointing a body axis somewhere ($\hat{b}\cdot\hat{d}$)

$\hat{d}_0 = \vec{R}(q)^T\hat{d}_\mathcal{E}$, $\cos(\psi) = \hat{b}^T\hat{d}^0$

Shortest rotation bringing $\hat{b}$ onto $\hat{d}^0$ turns by ψ about $\hat{b}\times\hat{d}^0$ (with len $\sin\psi$)

$$q_s = \left[\cos(\psi/2),\ \sin(\psi/2)\underbrace{\frac{\hat{b}\times\hat{d}^0}{\sin(\psi)}}_{\hat{n}}\right]$$ | $1+\cos\psi = 2\cos^2(\psi/2)$, $\sin\psi = 2\sin(\psi/2)\cos(\psi/2)$

$\cos(\psi/2) = \sqrt{\frac{\hat{b}^T\hat{d}^0+1}{2}}$

$\frac{\sin(\psi/2)}{\sin(\psi)} = \frac{1}{2\cos(\psi/2)} = \frac{1}{\sqrt{2(1+\hat{b}^T\hat{d}^0)}}$ $(\hat{b}\times\hat{d}^0)$

$$q_s = \begin{bmatrix}\frac{2\cos^2(\psi/2)}{2\cos(\psi/2)}\\ \frac{1}{2\cos(\psi/2)}(\hat{b}\times\hat{d}^0)\end{bmatrix} = \frac{1}{2\cos(\psi/2)}\begin{bmatrix}2\cos^2(\psi/2)\\ \hat{b}\times\hat{d}^0\end{bmatrix}$$

$$q_s = \frac{1}{\sqrt{2(1+\hat{b}^T\hat{d}^0)}}\begin{bmatrix}1+\hat{b}^T\hat{d}^0\\ \hat{b}\times\hat{d}^0\end{bmatrix}$$

and $q_d = q\otimes q_s$ $\Rightarrow q_e = q_d^*\otimes q = q_s^*\otimes q^*\otimes q$

$= q_s^*$

$\Rightarrow e_{0e} = \sqrt{\frac{1}{2}(1+\hat{b}^T\hat{d}^0)}$, (2ω), $\underbrace{e_p}_{\text{Perp}} = 2\vec{\varepsilon}_e = \frac{2\,\hat{d}^0\times\hat{b}}{\sqrt{2(1+\hat{b}^T\hat{d}^0)}}$

for small errs $\hat{d}^0 \approx \hat{b} - \delta\theta\times\hat{b}$ and denom $\to 2$

$\Rightarrow e_p \approx \hat{d}_0\times\hat{b} = -(\delta\omega\times\hat{b}) = \delta\theta - (\hat{b}\cdot\delta\theta)\hat{b} = \delta\theta_\perp$

Part of attitude error perp to $\hat{b}$

Integral + gains:

$u = -k_p e - k_i\int e\,dt - k_d\dot{e}$ and τ_d

$\Rightarrow J\ddot{e} + k_d\dot{e} + k_p e + k_i\int e\,dt = \tau_d$

$\frac{d}{dt}$ $\Rightarrow J\dddot{e} + k_d\ddot{e} + k_p\dot{e} + k_i e = 0$

and $Js^3 + k_d s^2 + k_p s + k_i$ Characteristic

Stable when $k_i > 0$ and $k_d k_p > J k_i$

add's small pole $-\alpha\omega_n$ ($\alpha \approx 0.1$–0.3, small)

$(s+\alpha\omega_n)(s^2 + 2\zeta\omega_n s + \omega_n^2) = s^3 + (2\zeta+\alpha)\omega_n s^2 + (1+2\zeta\alpha)\omega_n^2 s + \alpha\omega_n^3$

$\Rightarrow k_d = J(2\zeta+\alpha)\omega_n$, $k_p = J(1+2\zeta\alpha)\omega_n^2$ $k_i = J\alpha\omega_n^3$

Momentum and torque limits:

$\omega_{max,i} = \frac{\tau_{max}}{J_{ii}}$, and $|h_i| \approx J_{ii}|\omega_i| \le L_{max}$

$\Rightarrow |\omega_i| \le \frac{L_{max}}{J_{ii}}$

accelerating to α for half the slew and decelerating the other half, gives $\frac{1}{2}\alpha(t/2)^2 \Rightarrow \theta = \alpha t^2/8$

$t = 2\sqrt{\frac{\theta}{\alpha}}$, and $\omega_{peak} = \alpha\frac{t_{slew}}{2} = \sqrt{\alpha\theta}$

the wheel will saturate when $J\omega_{peak} > L_{max}$

so max slew that stays within mom. limit is

$\theta_{crit} = \frac{L_{max}^2}{J_{ii}^2\alpha_{max}}$

SINDy

Suppose there's a dynamical system with state vector $x(t) \in R^n$

$\Rightarrow \frac{d}{dt}[\vec{x}(t)] = \underline{f}(\underline{x}(t))$

eg $\frac{dV}{dt} = f(V(t)) = -\frac{V(t)}{RC}$

Assume f can be approximated as a linear combination of nonlinear functions Θ collected in a library matrix $\underline{\Theta}(\underline{X}) \in R^{m \times l}$, where $\underline{X} \in R^{m \times n}$ is the time series matrix over m measurements to be used in l candidate functions.

Usually start with simpler terms like polynomials of the state, then adding complexity

We want to find a coefficient matrix $\Xi \in R^{l \times n}$ s.t

$\dot{\underline{x}} \approx \Theta(\underline{x})\Xi$

so ie. $\underline{X} = \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \\ \vdots & \vdots \\ x_m & y_m \end{bmatrix} \Rightarrow \underline{\Theta}(\underline{X}) = \begin{bmatrix} | & | & | & | & | & | \\ 1 & x & y & x^2 & xy & y^2 \\ 1 & x_1 & y_1 & x_1^2 & x_1y_1 & y_1^2 \\ 1 & x_2 & y_2 & x_2^2 & x_2y_2 & y_2^2 \\ 1 & x_3 & y_3 & x_3^2 & x_3y_3 & y_3^2 \\ | & | & | & | & | & | \end{bmatrix}$ (m measurements $\times$ l functions)

then take $\Theta(\underline{X}) = [1 \;\; x \;\; y \;\; x^2 \;\; xy \;\; y^2]$ $(m \times l)$ (ie Θ takes a column of $\underline{X}$ through the function elementwise)

then $\Xi = \begin{bmatrix} | & | \\ \xi_1 & \xi_2 \\ 0 & 0.0 \\ 1 & 2.0 \\ \vdots & \vdots \end{bmatrix}$ (ξ_1 for $\dot{x}$, ξ_2 for $\dot{y}$) — l rows, how much each candidate function should be weighted; n columns

Higher degree terms are usually ill-conditioned, so measurement noise would wildly change coeff. values

$\Rightarrow$ quantified by $\kappa_2(\Theta)$ (big = bad)

Assuming sparsity in Ξ is the assumption, Brunton initially solves the ordinary least squares and then zeroes small values below λ

The regression:

$\min_{\xi_i} \|\dot{x}_i - \Theta(\underline{X})\xi_i\|^2$ (ie for column $\dot{x}_i$, so state variable i, solve to get corresponding ξ_i)

Too little data / limited initial conditions lead to this failing.

Coordinate basis selection is important (less work for the method)

So eg. SVD / DMD / autoencoder mapping to project to smaller latent, then SINDy on that latent to learn latent space dynamics.

POD: (truncate r modes)

$\underline{X} = \underline{U}\underline{\Sigma}\underline{V}^T \Rightarrow \underline{z} = \underline{U}_r^T \underline{X}$

DMD:

$\underline{Y} = [x(t_2) \;\; x(t_3) \;\; \dots] = \underline{A}\underline{X} = \underline{A}[\underline{x}(t_1) \;\; \underline{x}(t_2) \;\; \dots]$ and eigenvectors of A ($\underline{\Psi}$)

Can project $\underline{x}$ onto $\underline{z} = \underline{\Psi}^T \underline{x}$

Noise robustness is quite poor (i.e. also finite differences for derivatives)

$y = f(x) + \varepsilon(\sigma)$ (eg. $\varepsilon = A\sin(\omega_{noise} x)$)

$\Rightarrow \frac{d^n y}{dx^n} = f^n(x) + \begin{cases} (-1)^{\frac{n}{2}} A\omega_{noise}^n \sin(\omega_{noise} x) & \text{even} \\ \dots \end{cases}$ (so scales pretty bad)

Controller

$\frac{d}{dt}\underline{x}(t) = \underline{f}(\underline{x}(t), \underline{u}(t)) \Rightarrow \underline{\Theta}(\underline{x}, \underline{u})$ sum solve process

and can be used with eg MPC

Ensembles

train several on diff subsets, bootstrap, bagging

eg each of the m ensemble members is fitted independently

$\dot{\underline{x}}^{(a)} \sim \underline{\Theta}(\underline{x}^{(a)})\,\Xi^{(a)}$ then coefficients are compressed by their mean $\bar{\Xi} = \frac{1}{m}\sum_{a=1}^{m} \Xi^{a}$, will

Constrained Learning

We can define our own "educated" guesses for $\Theta(\underline{x})$

So eg for a drone $\underline{\Theta}(\underline{x}) = \begin{bmatrix} p, \tan\theta(q\sin\phi + r\cos\phi) \\ q\cos(\phi) - r\sin(\phi) \\ \vdots \end{bmatrix}^T$

and concatenate with control inputs eg $\underline{\Theta}(\underline{u}) = [u_1(t)\ u_2(t) \dots u_q(t)]$ (u_i is control input to i^{th} surface / control element)

$\Rightarrow \Theta(\underline{x}, \underline{u}) = [\underline{\Theta}(x)\ \underline{\Theta}(u)]$

$\Xi \in \mathbb{R}^{(r,1)}$ $r = 4\ (\phi)$ $n = 6$

we can then constrain with $\underline{C}$ $(\underline{C}\,\Xi(:) = \vec{d})$ $C \in \mathbb{R}^{n}$ $C = \begin{bmatrix} 1 & ? & & \\ 0 & 0 & 0 & \dots \\ 0 & 0 & 0 & \dots \\ 0 & 0 & 0 & \dots \end{bmatrix}$

So eg for $\dot{\phi} = 1$, we can use $C_{1,i} = \begin{cases} 1 & \text{if } j = 1 \\ 0 & \text{if } j \neq 1 \end{cases}$ with $d = 1$ ✓

and to enforce sparsity for this function for all the states $C_{i,j} = \begin{cases} 1 & \text{if } j = (i-1)(r+1) \\ 0 & \text{if } j \neq (i-1)(r+1) \end{cases}$ and $d_i = 0$ for $i \in [2, n]$